LEVEL I SCHWESER'S QuickSheet

CRITICAL CONCEPTS FOR THE 2027 CFA® EXAM

ETHICAL AND PROFESSIONAL STANDARDS

I Professionalism
I(A) Knowledge of the Law.
I(B) Independence and Objectivity.
I(C) Misrepresentation.
I(D) Misconduct.
I(E) Competence.
II Integrity of Capital Markets
II(A) Material Nonpublic Information.
II(B) Market Manipulation.
III Duties to Clients
III(A) Loyalty, Prudence, and Care.
III(B) Fair Dealing.
III(C) Suitability.
III(D) Performance Presentation.
III(E) Preservation of Confidentiality.
IV Duties to Employers
IV(A) Loyalty.
IV(B) Additional Compensation Arrangements.
IV(C) Responsibilities of Supervisors.
V Investment Analysis, Recommendations, and Actions
V(A) Diligence and Reasonable Basis.
V(B) Communication with Clients and Prospective Clients.
V(C) Record Retention.
VI Conflicts of Interest
VI(A) Avoid or Disclose Conflicts.
VI(B) Priority of Transactions.
VI(C) Referral Fees.
VII Responsibilities as a CFA Institute Member or CFA Candidate
VII(A) Conduct as Participants in CFA Institute Programs.
VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program.

QUANTITATIVE METHODS

Sources of Return

Price return: $\frac{P_1 - P_0}{P_0}$

Capital distribution return: $\frac{\text{Income}}{P_0}$

Total return = price return + distribution return

Multi-period Returns

Arithmetic mean return:

$$\frac{r_1 + r_2 + \ldots + r_n}{n}$$

Geometric mean return:

$$\sqrt[n]{(1+r_1)\times(1+r_2)\times\ldots\times(1+r_n)} - 1$$

Annualized return = $(1 + \text{HPR})^{365/\text{days}} - 1$

Continuous Compounding

$$r_{cc} = \ln(1 + \text{HPR}) = \ln\left(\frac{\text{end value}}{\text{beg value}}\right)$$

Security Market Indexes

Price weighting: Average of prices with denominator adjusted for splits.
Equal weighting: Equal weight on returns; requires rebalancing.
Market cap weighting: Weights are security market cap as proportion of index market cap.
Fundamental weighting: Weights based on firm fundamentals.

Time Value of Money

$$PV = \frac{FV}{(1 + r)^t}$$

With continuous compounding:

$$PV = \frac{FV}{e^{rt}}$$

Measures of Central Tendency

Arithmetic mean: Sum of all observations in sample/population, divided by number of observations.
Geometric mean: Used when calculating investment returns over multiple periods or to measure compound growth rates.
Trimmed mean (x%): Exclude highest and lowest x/2 percent of observations.
Winsorized mean (x%): Substitute values for highest and lowest x/2 percent of observations.
Median: Midpoint of dataset.
Mode: Most frequently occurring value.

Measures of Dispersion

Range = max value − min value

$$\text{Mean absolute deviation} = \frac{\sum_{i=1}^{n}|X_i - \bar{X}|}{n}$$

$$\text{Sample variance} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

Standard deviation = square root of variance

$$\text{Target semi-deviation} = \sqrt{\frac{\sum_{\text{all } X_i < B}^{n}(X_i - B)^2}{n-1}}$$

Skewness and Kurtosis

Positive (right) skew: mean > median > mode.
Negative (left) skew: mean < median < mode.
Leptokurtic (kurtosis > 3): More peaked (more observations close to mean) and fatter tails (more observations far from mean) than normal distribution.
Platykurtic (kurtosis < 3): Less peaked, thinner tails than normal distribution.

Covariance and Correlation

Covariance (population):

$$\sigma_{XY} = \sum_{i=1}^{n}\frac{(X_i - \mu_X)(Y_i - \mu_Y)}{n}$$

$$\text{Correlation} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Unconditional Mean and Variance

$$\mu_X = E(X) = \Sigma_i[x_i \times P(x_i)]$$
$$\sigma_X^2 = \text{Var}(X) = E[X - E(X)]^2$$
$$= E(X^2) - [E(X)]^2$$

Unconditional Covariance

$$\sigma_{XY} = \text{Cov}(X,Y)$$
$$= E\{[X - E(X)] \times [Y - E(Y)]\}$$
$$= E(XY) - E(X)E(Y)$$

Conditional Mean and Variance

$$E(X|S) = \sum_{i=1}^{n} X_i \times P(X_i|S)$$
$$\text{Var}(X|S) = E\left\{\left[X - E(X|S)\right]^2\right\}$$

Discrete Probability Distributions

Discrete uniform: Finite number of outcomes, each with equal probability.
Binomial: Success (probability p) or failure (probability $1 - p$); expected number successes in n trials = np; variance = $np(1 - p)$.
Poisson: Events happen independently at constant rate λ.

Continuous Probability Distributions

Continuous uniform: Infinite possible outcomes within range.
Normal: "Bell curve" defined by mean μ and standard deviation σ.
Standard normal: Normal with $\mu = 0$ and $\sigma = 1$.
Log-normal: Outcomes equal to e^x, where x follows a normal distribution.
Logistic: Resembles normal distribution but with fatter tails; used in logistic regression.

Bayes' Formula

$$P(A|B) = P(B|A)\frac{P(A)}{P(B)}$$

Central Limit Theorem

When selecting simple random samples of size n from population with mean μ and finite variance σ^2, the sampling distribution of the sample mean approaches a normal distribution with mean μ and variance σ^2/n as n becomes large.

Null and Alternative Hypotheses

Null hypothesis (H_0): Contains equal sign (=, ≥, ≤).
Alternative hypothesis (H_a): Concluded if evidence is sufficient to reject null hypothesis.

Type I and Type II Errors

Type I error: Rejecting null when true.
Type II error: Failing to reject null when false.

Hypothesis Tests

Test of:	*Stat:*	*d.f.*
Mean	t or z	$n - 1$
Difference in means	t	$n_1 + n_2 - 2$
Mean differences	t	$n - 1$
Variance	χ^2	$n - 1$
Equal variances	F	$n_1 - 1, n_2 - 1$
Correlation	t	$n - 2$
Independence	χ^2	$(r - 1)(c - 1)$
Regression slope: Significance Value	 F t	 $1, n - 2$ $n - 2$

Simulation Methods

Historical: Uses observed past data.
Bootstrap: Samples from past with replacement.
Monte Carlo: Uses assumed distributions of input data.

Linear Regression

$Y_i = b_0 + b_1 X_i + \varepsilon_i$
where:
Y = dependent variable
X = independent variable
b_0 = intercept term
b_1 = slope term
ε_i = error term (residual)
Estimated slope and intercept terms:

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

$$\hat{b}_1 = \frac{\text{Cov}_{XY}}{\sigma_X^2}$$

Analysis of Variance (ANOVA)

Total sum of squares (SST) = sum of squared differences between actual Y-values and $\bar{Y}$.
Sum of squares regression (SSR) = sum of squared distances between predicted Y-values and $\bar{Y}$.
Sum of squared errors (SSE) = sum of squared distances between actual and predicted Y-values.

Coefficient of Determination

R^2 = SSR/SST = percentage of variation in the dependent variable that is explained by variation in the independent variable.

ECONOMICS

Breakeven and Shutdown
Breakeven: total revenue = total cost.
Operate in short run if total revenue is greater than total variable cost but less than total cost.
Shut down in short run if total revenue is less than total variable cost.

Market Structures
Perfect competition: Many firms with no pricing power; very low or no barriers to entry; homogeneous product.
Monopolistic competition: Many firms; some pricing power; low barriers to entry; differentiated products; large advertising expense.
Oligopoly: Few firms that may have significant pricing power; high barriers to entry; products may be homogeneous or differentiated.
Monopoly: Single firm with significant pricing power; high barriers to entry; advertising used to compete with substitute products.
In *all* market structures, profit is maximized at the output quantity for which marginal revenue = marginal cost.

Savings, Investment, Fiscal Balance, and Trade Balance
Fiscal budget deficit (G − T) = excess of saving over domestic investment (S − I) − trade balance (X − M)

Business Cycle Phases
Expansion; peak; contraction; trough.

Economic Indicators
Leading: Turning points occur ahead of peaks and troughs.
Coincident: Turning points coincide with peaks and troughs.
Lagging: Turning points follow peaks and troughs.

Expansionary and Contractionary Policy
Monetary policy is expansionary when the policy rate is less than the neutral interest rate (real trend rate of economic growth + inflation target) and contractionary when the policy rate is greater than the neutral interest rate.
Fiscal policy is expansionary when a budget deficit is increasing or surplus is decreasing, and contractionary when a budget deficit is decreasing or surplus is increasing.

Balance of Payments
Current account: merchandise and services; income receipts; unilateral transfers.
Capital account: capital transfers; sales/purchases of nonfinancial assets.
Financial account: government-owned assets abroad; foreign-owned assets in the country.

Regional Trading Agreements
Free trade area: Removes barriers to goods and services trade among members.
Customs union: Members also adopt common trade policies with non-members.
Common market: Members also remove barriers to labor and capital movements among members.
Economic union: Members also establish common institutions and economic policy.
Monetary union: Members also adopt a common currency.

Foreign Exchange Rates
For the exam, FX rates are expressed as *price currency / base currency* and interpreted as the number of units of the price currency for each unit of the base currency.

Real Exchange Rate

$$\text{nominal FX rate} \times \left(\frac{\text{base currency CPI}}{\text{price currency CPI}}\right)$$

No-Arbitrage Forward Exchange Rate

$$\frac{\text{forward}}{\text{spot}} = \frac{1 + \text{price currency interest rate}}{1 + \text{base currency interest rate}}$$

Exchange Rate Regimes
Formal dollarization: Country adopts foreign currency.
Monetary union: Members adopt common currency.
Fixed peg: ±1% margin versus foreign currency or basket of currencies.
Target zone: Wider margin than fixed peg.
Crawling peg: Pegged exchange rate adjusted periodically.
Crawling bands: Width of margin increases over time.
Managed floating: Monetary authority acts to influence exchange rate but does not set a target.
Independently floating: Exchange rate is market-determined.

FINANCIAL STATEMENT ANALYSIS

Financial Statement Analysis Framework
1. State the objective and context
2. Gather data
3. Process the data
4. Analyze and interpret the data
5. Report conclusions or recommendations
6. Update the analysis

Auditor Opinions
Unqualified (unmodified, clean): Reasonable assurance that financial statements are free from material omissions and errors.
Qualified: Exceptions to accounting principles.
Adverse: Statements are not presented fairly or do not conform with accounting standards.
Disclaimer of opinion: Auditor is unable to express an opinion.

Revenue Recognition
Five-step revenue recognition model:
1. Identify contracts
2. Identify performance obligations
3. Determine transaction price
4. Allocate price to obligations
5. Recognize when (as) obligations are satisfied

Basic and Diluted EPS
Basic EPS calculation **does not** consider effects of any dilutive securities in computation of EPS:

$$\text{basic EPS} = \frac{\text{net income} - \text{preferred dividends}}{\text{wtd. avg. no. of common shs. outstanding}}$$

$$\text{diluted EPS} = \frac{\text{adj. income avail. for common shares}}{\text{wtd. avg. common shares plus potential common shares outstanding}}$$

Therefore, diluted EPS is:

$$\frac{\left[\begin{matrix}\text{net}\\\text{income}\end{matrix} - \begin{matrix}\text{pfd}\\\text{div}\end{matrix}\right] + \begin{matrix}\text{convertible}\\\text{preferred}\\\text{dividends}\end{matrix} + \left(\begin{matrix}\text{convertible}\\\text{debt}\\\text{interest}\end{matrix}\right)(1-t)}{\left(\begin{matrix}\text{wtd}\\\text{avg}\\\text{sh's}\end{matrix}\right) + \left(\begin{matrix}\text{shares from}\\\text{conversion of}\\\text{conv. pfd. sh's}\end{matrix}\right) + \left(\begin{matrix}\text{sh's from}\\\text{conversion of}\\\text{conv. debt}\end{matrix}\right) + \left(\begin{matrix}\text{shares}\\\text{issuable from}\\\text{stock options}\end{matrix}\right)}$$

Marketable Security Classifications
Held-for-trading: fair value on balance sheet; dividends, interest, realized and unrealized G/L recognized on income statement.
Available-for-sale: fair value on balance sheet; dividends, interest, realized G/L recognized on income statement; unrealized G/L is other comprehensive income.
Held-to-maturity: amortized cost on balance sheet; interest, realized G/L recognized on income statement.

Cash Flows From Operations (CFO)
Direct method: start with cash collections (cash equivalent of sales); cash inputs (cash equivalent of cost of goods sold); cash operating expenses; cash interest expense; cash taxes.
Indirect method: start with net income, subtracting gains and adding back losses resulting from financing or investment cash flows, adding back all noncash charges, and adding and subtracting asset and liability accounts that result from operations.

Free Cash Flow
Free cash flow (FCF) measures cash available for discretionary purposes and is equal to operating cash flow less net capital expenditures.

Critical Ratios
Common-size financial statement analysis:
- Common-size *balance sheet* expresses all balance sheet accounts as a percentage of total assets.
- Common-size *income statement* expresses all income statement items as a percentage of sales.
- Common-size *cash flow statement* expresses each line item as a percentage of total cash inflows (outflows), or as a percentage of net revenue.

Horizontal common-size financial statement analysis: expresses each line item relative to its value in a common base period.
Liquidity ratios:

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{quick ratio} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{current liabilities}}$$

$$\text{cash ratio} = \frac{\text{cash} + \text{marketable securities}}{\text{current liabilities}}$$

$$\text{defensive interval} = \frac{\text{cash} + \text{mkt. sec.} + \text{receivables}}{\text{daily cash expenditures}}$$

Receivables, inventory, payables turnover, and days' supply ratios—all of which are used in the cash conversion cycle:

$$\text{receivables turnover} = \frac{\text{annual sales}}{\text{average receivables}}$$

$$\text{inventory turnover} = \frac{\text{cost of goods sold}}{\text{average inventory}}$$

$$\text{payables turnover ratio} = \frac{\text{cost of goods sold}}{\text{average trade payables}}$$

$$\text{days of sales outstanding} = \frac{365}{\text{receivables turnover}}$$

$$\text{days of inventory on hand} = \frac{365}{\text{inventory turnover}}$$

$$\text{number of days of payables} = \frac{365}{\text{payables turnover ratio}}$$

$$\text{cash conversion cycle} = \left(\begin{matrix}\text{days of inventory}\\\text{on hand}\end{matrix}\right) + \left(\begin{matrix}\text{days of sales}\\\text{outstanding}\end{matrix}\right) - \left(\begin{matrix}\text{number of days}\\\text{of payables}\end{matrix}\right)$$

Total asset, fixed-asset, and *working capital turnover ratios:*

$$\text{total asset turnover} = \frac{\text{revenue}}{\text{average total assets}}$$

$$\text{fixed asset turnover} = \frac{\text{revenue}}{\text{average fixed assets}}$$

$$\text{working capital turnover} = \frac{\text{revenue}}{\text{average working capital}}$$

Gross, operating, and net profit margins:

$$\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}}$$

$$\text{operating profit margin} = \frac{\text{operating profit}}{\text{revenue}} = \frac{\text{EBIT}}{\text{net sales}}$$

$$\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$$

Return on assets [return on total capital (ROTC)]:

$$\text{return on assets (total capital)} = \frac{\text{EBIT}}{\text{average total capital}}$$

Debt to equity ratio and total debt ratio:

$$\text{debt-to-equity ratio} = \frac{\text{total debt}}{\text{total equity}}$$

$$\text{total-debt-ratio} = \frac{\text{total debt}}{\text{total assets}}$$

Interest coverage and fixed charge coverage:

$$\text{interest coverage} = \frac{\text{EBIT}}{\text{interest}}$$

$$\text{fixed charge coverage} = \frac{\text{EBIT} + \text{lease payments}}{\text{interest} + \text{lease payments}}$$

Growth rate: $g = RR \times ROE$

$$\textit{retention rate} = 1 - \frac{\text{dividends declared}}{\text{operating income after taxes}}$$

Liquidity ratios indicate company's ability to pay its short-term liabilities.
Operating performance ratios indicate how well management operates the business.

DuPont Analysis
Traditional DuPont equation:

$$\text{return on equity} = \left(\frac{\text{net income}}{\text{sales}}\right)\left(\frac{\text{sales}}{\text{assets}}\right)\left(\frac{\text{assets}}{\text{equity}}\right)$$

You may also see it presented as:

$$\text{return on equity} = \begin{pmatrix}\text{net profit}\\ \text{margin}\end{pmatrix}\begin{pmatrix}\text{asset}\\ \text{turnover}\end{pmatrix}\begin{pmatrix}\text{equity}\\ \text{multiplier}\end{pmatrix}$$

Extended DuPont equation further decomposes net profit margin:

$$ROE = \left(\frac{\text{net income}}{\text{EBT}}\right) \times \left(\frac{\text{EBT}}{\text{EBIT}}\right) \times \left(\frac{\text{EBIT}}{\text{revenue}}\right) \times \left(\frac{\text{revenue}}{\text{avg. total assets}}\right) \times \left(\frac{\text{avg. total assets}}{\text{avg. equity}}\right)$$

You may also see it presented as:

ROE = tax burden × interest burden × EBIT margin × asset turnover × leverage

Inventory Accounting
In periods of rising prices and stable or increasing inventory quantities:

LIFO results in:	*FIFO results in:*
Higher COGS	Lower COGS
Lower gross profit	Higher gross profit
Lower inventory balances	Higher inventory balances

Long-Lived Assets Capitalizing vs. Expensing
Capitalizing: lowers income variability and increases near-term profits. Increase assets, equity.
Expensing: opposite effect.

Revaluation of Long-Lived Assets
IFRS: revaluation gain recognized in net income only to the extent it reverses previously recognized impairment loss; further gains recognized in equity as revaluation surplus. (For *investment property*, all gains and losses from marking to fair value are recognized as income.)
US GAAP: revaluation is not permitted.

Deferred Taxes
- Created when taxable income (on tax return) ≠ pretax income (on financial statements) due to temporary differences.
- *Deferred tax liabilities* are created when taxable income < pretax income. Treat DTL as equity if not expected to reverse.
- *Deferred tax assets* are created when taxable income > pretax income. Must recognize *valuation allowance* if more likely than not that DTA will not be realized.

Leases
Lessee reporting: Under IFRS, lessee recognizes right-of-use asset (amortized straight-line) and liability equal to PV of lease payments. Interest portion of each payment is interest expense, principal portion decreases liability.
US GAAP is same except that right-of-use asset is amortized to match the lease liability, and for an operating lease, entire lease payment is an expense.
Lessor reporting, finance lease: Remove asset from balance sheet, recognize lease receivable asset, report interest income.
Lessor reporting, operating lease: Keep asset on balance sheet, report lease payments as income, record depreciation expense.

Pensions
Defined contribution: employer contribution expensed in period incurred.
Defined benefit: overfunded plan recognized as net pension asset, underfunded plan recognized as net pension liability.

Share-Based Compensation
For stock grants, performance shares, and employee stock options, the estimated fair value at the grant date is expensed over the vesting period.

Sales-Based Pro Forma Model
1. Estimate revenue growth
2. Estimate cost of sales
3. Estimate SG&A expenses
4. Estimate financing costs
5. Estimate taxes
6. Model working capital accounts
7. Estimate PP&E, capital spending
8. Construct pro forma statements

CORPORATE FINANCE

Corporate Governance
Stakeholder groups: Shareholders, board of directors, senior managers, employees, creditors, suppliers.
Key board committees: Audit, nominating/governance, compensation/remuneration.

Types of Capital Investment Projects
Going concern: Maintain the business or reduce costs.
Regulatory/compliance: Required to meet concerns such as safety or environment.
Expansion: Grow the existing business.
Other: Includes projects outside the existing business.

Capital Allocation
Administrative steps:
1. Generate ideas
2. Analyze project proposals
3. Create firm-wide capital budget
4. Monitor decisions and conduct post-audit

$$NPV = CF_0 + \frac{CF_1}{(1+k)^1} + \frac{CF_2}{(1+k)^2} + \ldots + \frac{CF_n}{(1+k)^n}$$

IRR: discount rate that makes NPV equal to zero.

Return on Invested Capital (ROIC)

$$ROIC = \frac{\text{net operating profit after tax}}{\text{average book value of invested capital}}$$

Real Options
Timing: Delay investment.
Abandonment: Exit project.
Expansion: Follow-on investment.
Flexibility: Change price or inputs.
Fundamental: Payoff depends on price of underlying asset.

Weighted Average Cost of Capital

$$WACC = (w_d)[k_d(1-t)] + (w_{ps})(k_{ps}) + (w_{ce})(k_s)$$

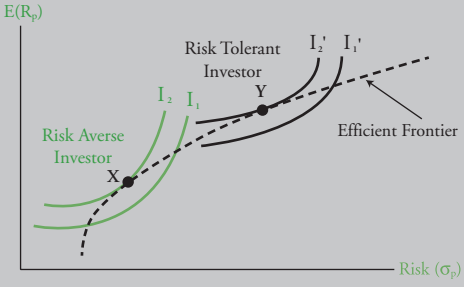
Capital Structure Theories
Modigliani-Miller (MM) with no taxes: Capital structure is irrelevant.
MM with taxes but no costs of financial distress: 100% debt maximizes firm value.
Static tradeoff theory: Firm value initially increases with debt financing, but decreases when costs of financial distress outweigh tax benefits of additional debt.
Pecking order theory: Managers prefer internal capital, then debt, then external equity.

PORTFOLIO CONSTRUCTION

Investment Policy Statement
Investment objectives:
- Return objectives.
- Risk tolerance.

Constraints:
- Liquidity needs.
- Time horizon.
- Tax concerns.
- Legal and regulatory factors.
- Unique circumstances.

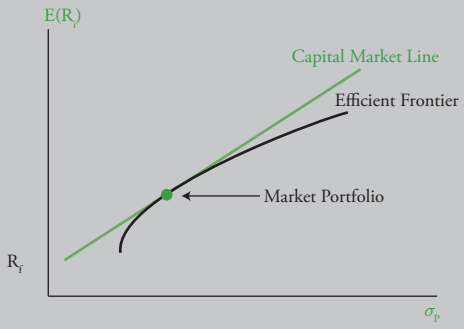
Combining Preferences with the Optimal Set of Portfolios
Markowitz efficient frontier is the set of portfolios that have highest return for given level of risk.



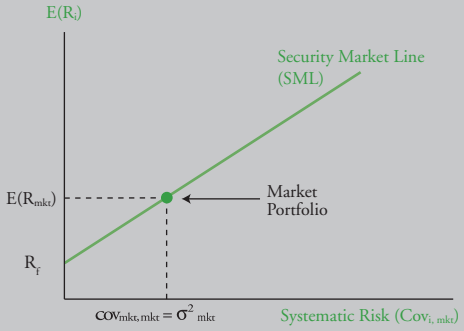
Security Market Line (SML)
Investors should only be compensated for risk relative to market. *Unsystematic risk* is diversified

away; investors are compensated for *systematic risk*. The equation of the SML is the CAPM, which is a return/systematic risk equilibrium relationship.

total risk = systematic + unsystematic risk



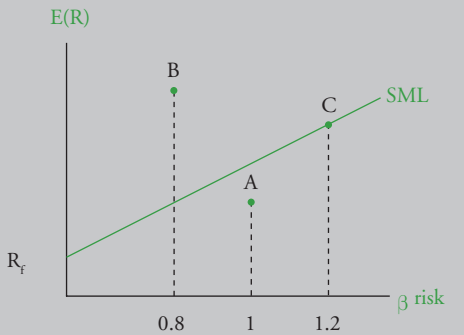
$$\text{CAPM: } E(R_i) = R_f + \beta_i[E(R_{mkt}) - R_f]$$



The SML and Equilibrium

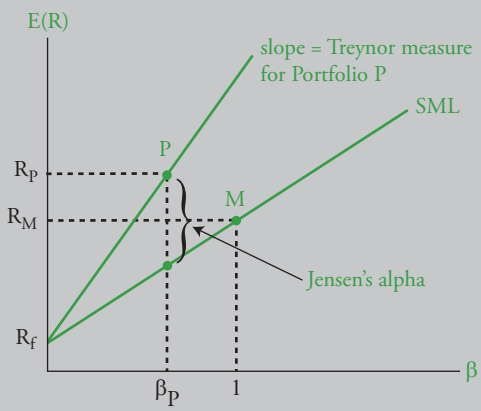
Identifying mispriced stocks:
Consider three stocks (A, B, C) and SML.
Estimated stock returns should plot on SML.

- A return plot over the line is *under*priced.
- A return plot under the line is *over*priced.



Risk-Adjusted Returns

Sharpe ratio and *M-squared* measure excess return per unit of *total risk*.
Treynor measure and *Jensen's alpha* measure excess return per unit of *systematic risk*.



Behavioral Biases

Cognitive errors, belief perseverance: Conservatism, confirmation, representativeness, control, hindsight.
Cognitive errors, information processing: Anchoring and adjustment, mental accounting, framing, availability.
Emotional biases: Loss aversion, overconfidence, self-control, status quo, endowment, regret aversion.

EQUITIES

Company Life Cycle

Start-up: No revenue, negative cash flow.
Growth: Increasing revenue, reinvestment of earnings, cash flow turning positive.
Mature: Stable cash flows, earnings distributed to equity owners.
Decline: Decreasing revenues and cash flows.

Equity Types

Common shares: Residual claim to earnings after debt and preference shares; typically has voting rights; may have multiple classes.
Preference (preferred) shares: Fixed payments as percentage of par value; may be callable, putable, cumulative, participating, or convertible to common shares.
Depository receipts: Used for indirect investing in foreign equity shares; may be sponsored (DR holder has voting rights) or unsponsored.

Primary and Secondary Markets

Primary market: Issuance of new securities (IPOs, private placements).
Secondary market: Trading of existing securities.

Bid and Ask Prices

Bid: Price at which a dealer will buy.
Ask (offer): Price at which dealer will sell.
Bid-ask spread is dealer's profit.

Market and Limit Orders

Market order: Execute immediately at best available price.
Limit order: Buy only at or below limit price; sell only at or above limit price.

Market Liquidity Measures

Market float: Total value of shares available to public (excludes controlling shares).
Average daily volume: Shares traded per day/number of days.
Turnover ratio: Avg. daily volume/market float.

Dividend Payment Chronology

Declaration: Board declares dividend, sets record and payment dates.
Ex-dividend: 1–2 business days before record date.
Record date: Date on which all owners of shares become entitled to receive the dividend payment.
Payment date: Dividend paid.

Indicators of Intrinsic Value

Book value of equity: Balance sheet assets minus liabilities and minority interest.
Market capitalization: Share price × shares outstanding.
Enterprise value: Market value of common and preferred equity + net debt (where net debt = market value of debt − cash and equivalents).

Equity Valuation Approaches

Absolute: Present value models (discounted cash flow, residual income), asset-based approach.
Relative: Multiplier models (e.g., P/E, P/BV, EV/EBITDA).

One-Period Valuation Model

$$V_0 = \frac{D_1}{1 - r_e} + \frac{P_1}{1 - r_e}$$

Infinite Period Dividend Discount Models

Constant growth model:

$$V_0 = \frac{D_1}{r_e - g_c} = \frac{D_0(1+g_c)}{r_e - g_c}$$

Supernormal growth (two-stage) DDM:

$$V_0 = \frac{D_1}{(1+r_e)} + \ldots + \frac{D_n}{(1+r_e)^n} + \frac{P_n}{(1+r_e)^n}$$

where: $P_n = \frac{D_{n+1}}{r_e - g_c}$

Free Cash Flow to Equity (FCFE)

FCFE = CFO − FCInv + net borrowing
FCFE = cash + dividends − (share issuance − share repurchases)
FCFE = net income − $\left(1 - \frac{\text{debt}}{\text{assets}}\right)$(net investment)

Free Cash Flow to the Firm (FCFF)

FCFF = CFO − capex + interest(1 − t)
FCFF = FCFE + interest(1 − t) − net borrowing
FCFF = [EBITDA(1 − t)] + (depreciation × t) − FCInv − WCInv

Residual Income

$RI_t = EPS_t - r \times BVPS_{t-1}$

Sustainable Growth Rate of Dividends

SGR = (1 − dividend payout ratio) × ROE

$$SGR = \left(1 - \frac{\text{FCFE}}{\text{net income}}\right)\left(\frac{\text{net income}}{\text{beginning book value}}\right)$$

$$SGR = \left(1 - \frac{\text{FCFF}}{\text{NOPAT}}\right)\left(\frac{\text{NOPAT}}{\text{beg. BV total capital}}\right)$$

where NOPAT = operating income(1 − t)

Justified Leading P/E Ratio

$$\frac{P_0}{E_1} = \frac{D_1 / E_1}{(r - g)} = \frac{(1 - b)}{(r - g)}$$

Justified Price/Book Value Ratio

$$\frac{P_0}{BVPS_0} = \frac{ROE - g}{r - g}$$

Justified Dividend Yield

$$\frac{D_0}{P_0} = \frac{(r - g)}{(1 + g)}$$

Industry and Competitive Analysis

1. Define the industry
2. Survey industry size, growth, profitability
3. Analyze industry structure (Porter)
4. Analyze external influences (PESTLE)
5. Analyze companies' strategies

Porter's Five Forces

1. Rivalry among existing competitors
2. Threat of entry
3. Threat of substitutes
4. Power of buyers
5. Power of suppliers

PESTLE Analysis

External forces affecting companies, industries:

- Political
- Economic
- Social
- Technological
- Legal
- Environmental

Competitive Strategies

- Cost leadership
- Product or service differentiation
- Focus (niche market)

FIXED INCOME

Price, Yield, Coupon Relationships

Bond prices and yields are inversely related.
Increase in yield decreases price; decrease in yield increases price.
Coupon < yield: Discount to par value.
Coupon > yield: Premium to par value.
Constant-yield price trajectory: Price approaches par as bond nears maturity from amortization of discounts and premiums. Capital gains and losses are calculated relative to this trajectory.
Other things equal, a *lower coupon rate* and a *longer maturity* make a bond price more sensitive to changes in yield.

Basic Features of Bonds

Issuer. Sovereign governments, corporations, local governments, agencies, supranational entities, special purpose entities.
Maturity. Money market (one year or less); capital market (greater than one year).
Par value. Bond's principal value (face value).
Coupon. Annual percent of par; fixed or floating. Divide by *periodicity* to get periodic rate.
Seniority. Senior > junior (subordinated).
Contingency provisions. Callable, putable, convertible.

Cash Flow Structures

Bullet: All principal repaid at maturity.
Fully amortizing: Equal periodic payments include both interest and principal.
Partially amortizing: Periodic payments include interest and principal, balloon payment at maturity repays remaining principal.
Sinking fund: Schedule for early redemption.
Floating-rate: Coupon payments based on reference rate plus margin.
Index-linked: Interest-indexed (coupon rate adjusted) or capital-indexed (principal adjusted).

Embedded Options

Callable: Issuer may repay principal early. Increases yield and decreases duration.
Putable: Bondholder may sell bond back to issuer. Decreases yield and duration.
Convertible: Bondholder may exchange bond for issuer's common stock.
Warrants: Bondholder may buy issuer's common stock at exercise price. Warrants are typically detachable from the bond.

Bond Markets

Domestic bonds. Domestic issuer and currency.
Foreign bonds. Foreign issuer, domestic currency.
Eurobond market is outside any one country, with bonds denominated in currencies other than those of countries in which bonds are sold.
Global bonds trade in both a national bond market and the eurobond market.

Bond Issuance

Underwritten offering: Investment banks buy entire issue, sell to public.
Best efforts offering: Investment banks act as brokers.
Shelf registration: Register entire issue with regulators but sell over a period of time.

Repurchase Agreements

Short-term borrowing with a fixed income security as collateral.
Repo rate: Annualized percent difference between sale and repurchase prices.
Initial margin: Extra collateral above loan amount.

$$Haircut = 1 - \frac{1}{\text{initial margin}}$$

Bond Pricing

Full price = PV on last coupon date $\times$

$$\left(1 + \frac{\text{YTM}}{\text{periods per year}}\right)^{\frac{\text{days since last coupon}}{\text{days in coupon period}}}$$

Accrued interest = coupon payment $\times$

$$\frac{\text{days from last coupon to settlement}}{\text{days in coupon period}}$$

Flat price = full price − accrued interest
Bonds are quoted at their flat prices.
Matrix pricing: For illiquid bonds, use yields of bonds with same credit quality to estimate yield; adjust for maturity differences with linear interpolation.

Yield Measures

Effective yield depends on periodicity. YTM = effective yield for annual-pay bonds.
Semiannual bond basis: YTM = 2 $\times$ semiannual discount rate.
Current yield = annual coupon / price.
Simple yield = current yield ± amortization.
Yield to call is based on call date and call price.
Yield to worst is lowest of a bond's YTCs or YTM.

Yield Spreads

G-spread: Basis points above government yield.
I-spread: Basis points above swap rate.
Z-spread: Accounts for shape of yield curve.
Option-adjusted spread: Adjusts Z-spread to remove effects of embedded options.

Floating-Rate Notes

Quoted margin: Fixed margin above the MRR.
Required margin or *discount margin:* Margin above the MRR that would price the note at par. Decrease in credit quality causes the required margin to be greater than the quoted margin.

Money market yields may be on a discount or add-on basis and may use a 360- or 365-day year. *Bond-equivalent yield* is an annualized add-on yield based on a 365-day year.

Forward and Spot Rates

Forward rate is a rate for a loan that begins at a future date. "1y3y" = 3-year forward rate 1 year from today.
Example of spot-forward relationship:

$$(1 + S_2)^2 = (1 + S_1)(1 + 1y1y)$$

Interest Rate Risk

Interest rate risk has two components: *reinvestment risk* and *market price risk* from YTM changes. These risks have opposing effects on an investor's horizon yield.

- Bond investors with short horizons are more concerned with market price risk.
- Bond investors with long horizons are more concerned with reinvestment risk.
- The horizon at which market price risk and reinvestment risk just offset is a bond's *Macaulay duration.* This is the weighted average of times until a bond's cash flows are scheduled to be paid.

Modified duration is the approximate change in a bond's price given a 1% change in its YTM:

$$= \frac{\text{Macaulay duration}}{(1 + r)} \approx \frac{(V_-) - (V_+)}{2V_0(\Delta y)}$$

Price change estimates based on duration only are improved by adjusting for *convexity:*

$$\%\Delta\text{price} = -\text{duration}(\Delta y) + \frac{1}{2}\text{convexity}(\Delta y)^2$$

Approximate convexity:

$$\frac{V_- + V_+ - 2V_0}{(\Delta\text{YTM})^2 V_0}$$

Effective duration is required if a bond has embedded options:

$$\frac{(V_-) - (V_+)}{2V_0(\Delta\text{curve})}$$

Credit Risk and Analysis

Bottom-up credit analysis factors:

- Capacity
- Capital
- Collateral
- Covenants
- Character

Top-down credit analysis factors:

- Conditions
- Country
- Currency

Expected loss = probability of default $\times$ loss given default

Credit ratings:
Investment grade: Baa3/BBB− or above.
Non-investment grade: Ba1/BB+ or below.
Corporate family rating (CFR): issuer rating.
Corporate credit rating (CCR): security rating.

Secured bonds are backed by specific collateral and senior to unsecured bonds.
Unsecured bonds are general claims to issuer's cash flows and assets.

Asset-Backed Securities

Residential MBS: Home mortgages are collateral.
Agency RMBS include only conforming loans; nonagency RMBS may include nonconforming loans and need credit enhancement.
Internal credit enhancement: Excess spread, overcollateralization, waterfall structure.
Prepayment risk: Contraction risk from faster prepayments; extension risk from slower prepayments.
CMOs: Pass-through MBS are collateral. May have sequential-pay or PAC/support structure.
Commercial MBS: Non-recourse mortgages on commercial properties are collateral.
Credit card ABS: Credit card receivables are collateral.
CDOs: Bonds, bank loans are collateral; employ a collateral manager.

DERIVATIVES

Arbitrage and Replication

- *Law of one price*: two assets with identical cash flows in the future, regardless of future events, should have the same price.
- Two assets with uncertain returns can be combined in a portfolio that will have a certain payoff. If a portfolio has a certain payoff, the portfolio should yield the risk-free rate. For this reason, derivatives values are based on risk-neutral pricing.

Derivatives Values vs. Prices

The price of a forward, futures, or swap contract is the forward price stated in the contract and is set such that the contract has a *value* of zero at initiation. Value may change during the contract's life with opposite gains/losses to the long and short.

Forward Contract Value

At time *t*:

$$V_t(T) = [S_t + PV_t(\text{costs}) - PV_t(\text{benefits})] - F_0(T)(1 + Rf)^{-(T-t)}$$

Futures vs. Forwards
Futures are standardized, exchange-traded forward contracts that require daily cash settlement of mark-to-market gains and losses.

Forward Rate Agreements (FRA)
Can be viewed as a forward contract to borrow/lend money at a certain rate at some future date.

Interest Rate Swaps
May be replicated by a series of FRAs with present values at swap initiation that sum to zero.

Options
- Buyer of a call option—long asset exposure.
- Writer (seller) of a call option—short asset exposure.
- Buyer of a put option—short asset exposure.
- Writer (seller) of a put option—long asset exposure.

intrinsic value of a call option = $\text{Max}[0, S - X]$
intrinsic value of a put option = $\text{Max}[0, X - S]$

American vs. European Options
American options allow the owner to exercise the option any time before or at expiration. *European options* can be exercised *only* at expiration.

Factors that Affect Option Values

Increase in:	*Calls*	*Puts*
Asset price	Increase	Decrease
Exercise price	Decrease	Increase
Risk-free rate	Increase	Decrease
Volatility	Increase	Increase
Time to expiration	Increase	Increase*
Holding costs	Increase	Decrease
Holding benefits	Decrease	Increase

*Except some deep-in-the-money European puts.

Put-Call Parity
The put-call parity relationship for European options at time *t*:

$$c + X(1 + Rf)^{-T} = S + p$$

Each security in the put-call parity relationship can be expressed as:

$$S = c - p + X(1 + Rf)^{-T}$$
$$p = c - S + X(1 + Rf)^{-T}$$
$$c = S + p - X(1 + Rf)^{-T}$$
$$X(1 + Rf)^{-T} = S + p - c$$

Put-Call Forward Parity
The present value of the forward price of the underlying asset, $F_0(T)(1 + Rf)^{-T}$, can be substituted for S in any of the put-call parity relationships at time 0.

ALTERNATIVE INVESTMENTS

Alternative Investment Life Cycle
Capital commitment phase: Managers identify investments and make capital calls from partners.
Capital deployment phase: Managers fund and involve themselves in projects.
Capital distribution phase: Successful investments generate income and cash flows.

Fair Value Hierarchy
Level 1: Active market, quoted prices readily available.
Level 2: Can be valued based on observable inputs or with models.
Level 3: Require unobservable inputs to establish a fair value; few or no market transactions.

Redemption Restrictions
Lockup period: Time after initial investment during which limited partners cannot request redemptions or incur significant penalties.
Notice period: Time within which a fund must fulfill a redemption (typically 30–90 days).
Gate: Temporary restriction of redemptions.

Fee Structures
- *Management fee and performance fee,* for example "2 and 20": 2% management fee plus 20% performance fee. Management fee is typically a percentage of assets under management for hedge funds, percentage of committed capital for private equity.
- *Hard hurdle rate:* incentive fee only on return above hurdle rate.
- *Soft hurdle rate:* incentive fee on whole return, but only paid if return is greater than hurdle rate.
- *High water mark:* no incentive fee until value exceeds previous high.
- *Clawback provision:* limited partners can recover performance fees if earlier gains are reversed.
- *Deal-by-deal (American) waterfall:* profits distributed as each investment is sold.
- *Whole-of-fund (European) waterfall:* limited partners receive all distributions until they have recovered their initial investment plus the hurdle rate.

Hedge Funds
Event-driven strategies: merger arbitrage; distressed/restructuring; activist shareholder; special situations.
Relative value strategies: convertible arbitrage; specific fixed income; general fixed income; multi-strategy.
Equity strategies: market neutral; fundamental growth; fundamental value; fundamental long/short; short bias.
Opportunistic strategies: macro; managed futures.

Private Capital
Leveraged buyouts: management buyouts (existing managers), management buy-ins (new managers)
Venture capital stages of development:
- Formative stage: angel investing, seed stage, early stage.
- Later stage: expand production, increase sales.
- Mezzanine stage: prepare for IPO.

Exit strategies: trade sale; IPO; recapitalization; secondary sale; write-off.

Real Estate
Includes residential property; commercial property; real estate investment trusts (REITs); whole loans; construction loans.

Commodities
Contango: futures price > spot price.
Backwardation: futures price < spot price.

$$\text{futures price} \approx \text{spot price}(1 + R_f) + \text{storage costs} - \text{convenience yield}$$

Infrastructure
Long-lived assets for public use, including transportation, utility, communications, social.
Greenfield: Invest in new infrastructure assets, often on a "Build-Operate-Transfer" basis.
Brownfield: Develop existing infrastructure assets.
Second-stage: Invest in infrastructure assets that do not need further development.

ISBN: 978-1-0788-5650-8
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